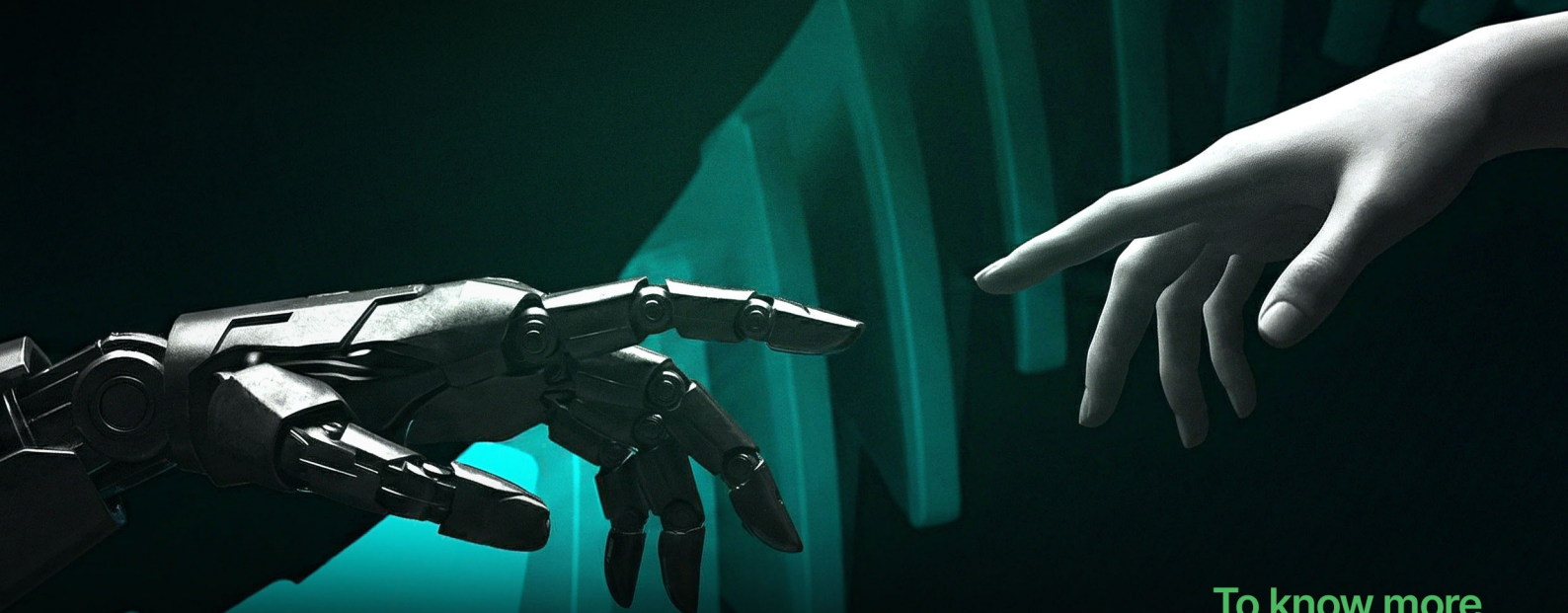


AI-DRIVEN SOFTWARE ENGINEERING BOOTCAMP

BATCH 07



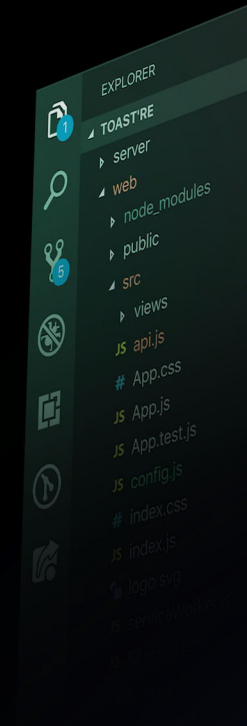
PLACEMENT STATISTICS

Full Stack Developer	54%
Frontend Developer	22%
Backend Developer	18%
Others	6%

Full time	68%
Paid Intern	24%
Contractual	8%

Onsite	53%
Remote	42%
Hybrid	5%

IT	59%
Non-IT	41%



OUR MISSION

The core purpose of Next Level is to take students beyond the basics and equip them with industry-standard, advanced skills so they can truly compete in the global job market. While many learners already have foundational knowledge, they often struggle to grow in advanced technologies due to a lack of structured guidance and consistent support. This is where Next Level makes the difference. Our Mentor Team continuously researches the latest tools, frameworks, and best practices, designing a curriculum that stays aligned with current trends and future industry demands. We don't aim to simply deliver content; we aim to bridge the gap between learning and real-world requirements by preparing students to work on complex, real-world projects, gain proficiency in tools used by top global companies, and become capable full-stack developers who can confidently handle both frontend and backend. At the same time, we recognize a major shift in the industry toward AI-driven development, which is why we have strategically integrated AI-focused components into our curriculum including AI Fundamentals, Retrieval-Augmented Generation (RAG), Vector Databases, Semantic Search, LangChain, AI Model Integration, and AI Agents with Automation. This ensures that students are not just writing code, but building intelligent, future-ready applications and systems. Ultimately, our mission is to ensure that what students learn doesn't stop at knowledge, but leads them all the way to real career opportunities helping them stand out, compete globally, and represent Bangladesh as creators of impactful, next-generation technology.

Looking ahead, our vision is to create a new generation of engineers who are not only job-ready but capable of building world-class products and solutions that make a global impact. We aim to transform Bangladesh into a strong hub of tech talent where developers can confidently work in international teams, contribute to global products, and unlock massive remote career opportunities.

To know more

 level2@programming-hero.com



01322-810873

NEXT LEVEL AI-DRIVEN

SOFTWARE ENGINEERING ROADMAP

01

BE A TYPESCRIPT TECHNOCRAT

MODULE 01 / TYPESCRIPT FOUNDATIONS

- TypeScript setup
 - Installing Node.js using NVM
 - Basic data types in TypeScript
 - Arrays and objects
 - Functions in TypeScript
 - Optional types
 - Literal types
 - Spread operator
 - Rest operator
 - Destructuring
 - Type alias
 - Union and intersection types
 - Ternary, nullish coalescing, and optional chaining
 - Nullable, unknown, and never types
-

MODULE 02 / ADVANCED TYPESCRIPT TYPES

- Type assertion
- Interfaces
- Generics
- Generics with interfaces
- Generics with functions
- Constraints in TypeScript
- KeyOf constraint with generics
- Enum
- Using as const instead of Enum
- Conditional types
- Mapped types
- Utility types

MODULE 03 / OBJECT-ORIENTED PROGRAMMING IN TYPESCRIPT

- Classes and objects
- Inheritance (1st pillar of OOP)
- Type guards using typeof and in
- Type guards using instanceof
- Access modifiers
- Getters and setters
- Static properties and methods
- Polymorphism (2nd pillar of OOP)
- Abstraction (3rd pillar of OOP)
- Encapsulation (4th pillar of OOP)

MODULE 04 / ASSIGNMENT 1

- Use TypeScript to solve real programming problems
- Work with types, interfaces, arrays, classes, generics, and functions

02

BE A NODE, EXPRESS
& SQL SUPERSTAR

MODULE 05 / FUNDAMENTALS OF WEB WITH NODE.JS

- How the Web Works from Scratch
 - Frontend vs Backend Essentials
 - Tracing Where Everything Began
-

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- How Node.js Revolutionized the Server Landscape
 - Event-Driven Architecture of Node.js
 - Single-Threaded Node.js vs Multithreaded Servers
 - Streams, Buffers & Modular Systems Deep Dive
 - CommonJS Require/Exports, ESM Transition
 - Harnessing IIFE as the Ultimate Module Wrapper
 - Module Summary and Key Takeaways
-

MODULE 06 / BUILDING SERVER WITH RAW NODE.JS & TYPESCRIPT

- Building Server with Node.js
 - Route Handling with HTTP Requests & Responses
 - Better File Organizations
 - Reading Files With fs module
 - Building GET Method Operations
 - Parsing Request Body
 - Creating clean POST API
 - Perfecting PUT Method Implementation
 - Executing DELETE Method Operations
 - Optimizing sendResponse
 - Working with Environment-based Configurations
 - Module Summary
-

MODULE 07 / CRUD OPERATION WITH EXPRESS, TYPESCRIPT & POSTGRESQL

- Create server with Express & TypeScript
 - Understanding the Express Request and Response
 - Setting Up Postgres with Neon Serverless Cloud
 - Explore SQL Data types with practical implementations
 - Executing Pool and creating tables
 - Creating our first User with POST method
 - Getting All Users and Single user with params
 - Update User with the PUT method
 - Delete User with Delete Method
 - Set up Environment-based Configurations
 - Module Summary
-

MODULE 08 / MODULAR PATTERN & AUTHENTICATION FLOW

- Introducing to the Modular pattern
 - Cleaning up the Server and creating app.ts
 - Implementing the modular pattern
 - Refactoring the User modules
 - Managing the user profile with SQL reference
-

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- Creating User Profiles Management
 - Understanding Authentication and Authorization
 - Introduction with Bcrypt and hashing password
 - Theory of JWT and Generate Token
 - Module Summary
-

MODULE 09 / AUTHORIZATION, REFRESH TOKEN & GLOBAL ERROR HANDLING

- Understanding the Middleware
- Higher order functions with auth middleware
- Auth middleware & Declaring user in Namespace
- Role Permission System Implementation
- Implementing Business logic for User status
- Secure Refresh Token API
- Global Error Handling Architecture
- SendResponse Utility Creation
- Complete Workflow Review
- Module Summary

MODULE 10 / ASSIGNMENT 2

- Build a basic backend server using Node.js and Express
- Connect to a SQL database
- Create RESTful APIs
- Implement authentication and authorization

03

BE A RDBMS
ROCKSTAR

MODULE 11 / FUNDAMENTALS OF DATABASES AND DATA MODELING

- Database and data vs information
 - Why file systems fail for modern applications
 - Types of database models
 - The anatomy of a table or relation
 - Keys in databases: super key
 - Candidate keys, subsets & proper subsets
 - Primary, alternate, simple, and composite keys
 - Foreign keys explained
 - Database design process: step-by-step
 - Relationship cardinality and ER diagrams
-

MODULE 12 / DATABASE NORMALIZATION TECHNIQUES

- Data anomalies and types of anomalies in DBMS
- Normalization and functional dependency
- 1NF, 2NF, 3NF explained with examples
- Resolving many-to-many relationships
- Updating ER diagram using a junction table
- What is PostgreSQL?
- Install PostgreSQL and explore psql CLI
- Some PostgreSQL commands — add psql to PATH

MODULE 13 / HANDS-ON SQL & POSTGRESQL BASICS

- PgAdmin basics
- Install Beekeeper Studio
- Integer and Boolean data types
- Character, Date, and UUID data types
- Create and drop database / table
- Column constraints
- Multiple constraints and INSERT
- Insert data methods
- Insert without column names

MODULE 14 / POSTGRESQL ADVANCED DATA MANIPULATION TECHNIQUES

- ALTER table and constraints
- ALTER default values and column constraints
- SELECT basics: sorting and aliases
- DISTINCT and WHERE filtering
- Filtering with AND and OR
- Comparison, BETWEEN, and IN
- LIKE vs ILIKE
- NOT and scalar functions
- Aggregate functions explained

MODULE 15 / POSTGRESQL ESSENTIALS — DATA, KEYS, AND JOINS

- Handling NULL with COALESCE
 - LIMIT, OFFSET, pagination
 - Updating and deleting data
 - GROUP BY and GROUP BY with HAVING
 - Foreign key explained
 - Understanding INNER JOIN
 - LEFT JOIN, RIGHT and FULL JOIN
 - All joins overview (including CROSS and NATURAL join)
-

MODULE 16 / ASSIGNMENT 3

- Apply SQL fundamentals and design databases
- Understand core database concepts
- Skills essential for Senior Developer projects

04

BE A PRISMA PRO & NEXT.JS NINJA

MODULE 17 / WELCOME TO PRISMA ORM

- Prisma learning plan
- Setting up Prisma and database connection
- Your first Prisma query and migration
- Writing Prisma schema and model relations
- Prisma model relations and enums
- Creating, reading, updating and deleting records
- Upsert in Prisma

MODULE 18 / REQUIREMENT ANALYSIS, ERD & PROJECT SETUP

- Requirement analysis of the blog application
- High-level system design for the blog app
- Designing the ER diagram for the blog app
- Understanding entity relationships in the ER diagram
- Project initialization and setup
- Defining Post and Comment models using Prisma schema
- Indexing, mapping, and self-relations in Prisma schema
- Express server and application configuration
- Implementing POST API

MODULE 19 / EMAIL & PASSWORD AUTHENTICATION WITH CUSTOM AUTH

- Email & password authentication setup
 - Extending the User model: adding additional fields
 - Email verification system (token and flow design)
 - Email verification system (email sending and validation)
 - Social authentication: sign up & sign in with Google
 - Advanced authentication configuration and best practices
 - Authentication middleware — basic protection
 - Authentication middleware — role-based access control
 - Authentication middleware — production-ready security
-

MODULE 20 / ADVANCED QUERYING, FILTERING

- Understanding CORS
- Designing a better authentication system
- Fetching all posts from the database
- Searching posts by title
- Advanced search using OR and HAS conditions
- Filtering posts by tags, boolean fields, status and author ID

MODULE 21 / ADMIN SEEDING, DATA PAGINATION & SORTING

- Admin seeding
- Offset pagination and cursor-based pagination
- Calculation and helpers for pagination and sorting
- Retrieve post data with pagination metadata
- Get post by ID
- Increment view count and transaction

MODULE 22 / COMMENT SYSTEM IMPLEMENTATION

- Set up comment routes in Express
- Design comment models with post and author relationships
- Handle nested replies
- Implement comment count and ordering
- Query comments efficiently by post ID or author ID
- Implement CRUD operations with cascade delete
- Retrieve comments with related post data

MODULE 23 / ADMIN FEATURES, POST MANAGEMENT & ANALYTICS

- Admin comment status management
- Retrieve my posts — advanced filtering and pagination
- Update and delete user-owned posts
- Role-based post update access
- Statistics & analytics API
- Multi-file schema architecture

MODULE 24 / GLOBAL ERROR HANDLING & PRISMA ERROR MANAGEMENT

- Global error handling middleware
 - Handling Prisma client validation errors
 - Managing Prisma client known request errors
 - Dealing with unknown and initialization errors
 - Not Found (404) middleware implementation
 - Module summary and key takeaways
-

MODULE 25 / INTRODUCTION TO NEXT.JS

- Why Next.js — React and SPA limitations
- Key Next.js features
- App Router vs Page Router and bundlers
- Setting up your project and integrating Shadcn/UI
- File-system routing basics
- Creating basic routes and understanding core layout concepts
- Mastering layouts and Global UI
- Navigations in Next.js

MODULE 26 / ADVANCED NEXT.JS LAYOUTS, LOADING, AND ROUTING

- Dark mode and hydration errors
- loading.js and error.js files
- Static vs dynamic builds
- Implementing multiple layouts using route groups
- Understanding and implementing parallel routes
- Role-based layout and sidebar
- Type definition for sidebar and module summary

MODULE 27 / AUTHENTICATION DEEP DIVE — TOKEN, SESSION & COOKIES

- AAA security framework
- Token-based vs session-based authentication
- Cookies vs localStorage
- Login and registration page with custom auth setup
- Authentication with Custom Auth
- Cookies in Next.js client and server components

MODULE 28 / DATA FETCHING, SESSIONS, AND ENVIRONMENT SETUP

- Get session data in server components
- Service layer creation
- Environment variables with type-safe validation
- Private routes using proxy.ts
- Caching behaviors: SSG, ISR, and SSR
- Handling params and fetch options

MODULE 29 / BUILDING POST FORMS, CACHE REVALIDATION & PAGINATION

- Creating server form component with server action
 - On-demand cache revalidation
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- Handling server action using client component
 - Creating table using Shadcn
 - Handling URL searchParams
 - Bootstrapping PaginationControls component
 - Parallel data fetching and Next.js Image component
-

MODULE 30 / ASSIGNMENT 4

- Full-stack industry-standard project built without AI assistance
- Evaluate skills and develop critical thinking from scratch

05

BE A BACKEND BRAINIAC

MODULE 31 / REQUIREMENT ANALYSIS AND PROJECT SETUP

- Project Requirement Analysis
- Project Database ERD Analysis
- Setting Up The Project
- Connecting Prisma With The Project And Creating Specialty Model
- Creating Route, Controller, Service For Speciality Model API
- Completing Specialties API
- Creating Send Response Utility Functions
- Setting Up Multi File Prisma Schema
- Setting Up Better Auth And Creating Auth Schemas
- Registering First User As Patient
- Login User With Email Password

MODULE 32 / DOCTOR MANAGEMENT AND JWT AUTHENTICATION

- Environment Variables & Create Patient Model
- Completing Patient Registration With Transaction
- Creating Global Error Handler And Not Found
- Completing Create Doctor & Doctor Specialty Model
- Creating Get All Doctor Route
- Introducing Zod Validation Library
- Creating Zod Schema, Middleware and Error Handler
- Create Custom Error Class And Handle Native Errors
- Creating Access Token And Refresh Token With JWT
- Create Authorization Middleware And Protecting Routes

MODULE 33 / ADVANCED AUTHENTICATION FEATURES

- Add Custom Property Req.User And Creating /Me Route
 - Getting New Access Token With Refresh Token
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- Creating Change Password And Logout API
 - Creating Email Sender With Nodemailer And Google Cloud
 - Setting Up Email Verification With OTP
 - Setting Up Reset Password With OTP
 - Creating Google Login API
 - Setting Up CORS And Better Auth Trusted Origins
-

MODULE 34 / FILE UPLOAD AND QUERY BUILDER

- Introduction To Multer And Cloudinary For Image Upload
 - Refactoring Create Specialty API For Image Upload
 - Creating Function To Upload And Delete File From Cloudinary
 - Introducing Query Builder
 - Creating Search, Filter, Paginate, Sort, Fields Selection Methods
 - Creating Include And Dynamic Include Methods
 - Implementing Query Builder In Get All Doctors API
 - Refactoring Query Builder And Fixing Issues
-

MODULE 35 / SCHEDULING AND PAYMENT INTEGRATION

- Creating Create Schedule API
 - Completing Schedule APIs And Creating Doctor Schedule APIs
 - Creating Book Appointment API
 - Setting Up Stripe Payment System
 - Creating Payment Webhook Handler
 - Auto-Cancelling Unpaid Appointments With Cron Job
 - Seed Super Admin During Starting The Server
 - Refactoring And Fixing Book Appointment Route And Payment System
-

MODULE 36 / PROFILE UPDATES AND PDF GENERATION

- Update My Profile Route For Patient Role
 - Creating Review APIs
 - Completing Prescription APIs
 - Prescription PDF Generation And Uploading To Cloudinary
 - Creating Email Template For Prescription
 - Payment Invoice PDF Generation And Sending Payment Invoice Email
-

MODULE 37 / ADMIN DASHBOARD AND ERROR HANDLING

- Creating Dashboard API
 - Creating Dashboard Bar Chart And Pie Chart Functions
 - Creating Change User Status API And Change User Role API
 - Analysing Prisma Error Handling Patterns And Error Codes
 - Creating Handlers For Prisma Client Errors
 - Handle Uncaught Exception, Unhandled Rejection, SIGTERM, SIGINT Errors
-

MODULE 38 / INDUSTRY STANDARD PROJECT SETUP

- Initializing The Frontend With NextJS 16
- Planning The Routing Architecture Of PH Healthcare Frontend
- Routing Setup For Public Routes
- Routing Setup For Private Routes
- Layout Setup In The Project
- Adding Not Found Page, Error Page, Loading Pages And ShadCN Components In The Project
- Setting Up Axios HTTP Client For Data Fetching
- Introduction To Tanstack Query
- Setting Up Tanstack Query Provider In Server Side Of NextJs
- Tanstack Query Prefetch And Dehydrate Technique For Data Fetching In Server Side Of NextJs

MODULE 39 / ADVANCE LOGIN FORM IMPLEMENTATION

- Refactoring Http Client Methods of Axios And Adding API Types
- Creating Zod Validation Schema, Auth Types For Login
- Creating Server Action For Login
- Setting Access Token, Refresh Token, Better Auth Tokens In Cookies From Server Action With NextJS Cook
- Creating Reusable Form Input Field Component
- Creating Reusable Form Submit Button
- Creating Login Form Component UI With Tanstack Form, Tanstack Query And Zod Part-1
- Creating Login Form Component UI With Tanstack Form, Tanstack Query And Zod Part-2
- Creating Login Form Component UI With Tanstack Form, Tanstack Query And Zod Part-3
- Testing Login Functionality From Browser

MODULE 40 / SECURE AUTHENTICATION SYSTEM WITH REFRESH**TOKENS AND ROUTE GUARDS**

- Setting Up Proxy File And Planning On Route Protection And Authorization
 - Creating Authenticated And Authorized Routes List With Helper Functions Auth Utilities
 - Protecting And Authorizing The Pages With Main Proxy Functions
 - Redirecting Unauthenticated Users To Target Route After Login.Mp4
 - Creating Server Action For Refresh Token And Planning How New Access Token Will Set
 - Setting New Access Token In Frontend Using Refresh Token Inside Proxy File
 - Setting New Access Token In Frontend Using Refresh Token With Axios
 - Protecting Reset Password Route For Need Password Change
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- Enforcing Users To Change Password With Need Password Change Boolean Value
 - Enforcing Users To Verify Email With Email Verified Boolean Value
-

MODULE 41 / CREATING DASHBOARD COMPONENTS

- Setting Up Dashboard Layout Structure
 - Creating Dynamic Role Based Navigation Items For Sidebar
 - Making Dashboard Sidebar Component Part-1
 - Making Dashboard Sidebar Component Part-2
 - Creating Icon Mapper Function To Get Dynamic Icons For Sidebar
 - Making Dashboard Navbar Component Part-1
 - Making Dashboard Navbar Component Part-2
 - Making Dashboard Navbar Component Part-3
 - Making Dashboard Navbar Component Part-4
 - Making Dashboard Navbar Component Part-5
 - Making Dashboard Navbar Component Part-6
 - Making Dashboard Navbar Component Part-7
-

MODULE 42 / DASHBOARD DATA VISUALIZATION & TABLES

- Creating Stats Card Component For Dashboard
 - Creating Pie Chart Component For Dashboard
 - Creating Bar Chart Component For Dashboard
 - Understanding How Dashboard Page Will Be Developed
 - Creating Dashboard Data Interface, Server Action And Prefetching With Tanstack Query
 - Completing Dashboard Page With UseQuery Hook And Dashboard Components
 - Introduction To Tanstack Table
 - Setting Up Server Action And Interface To Get All Doctors
 - Completing Doctors Table With Tanstack Table
-

MODULE 43 / ADVANCED TABLE COMPONENTS & CUSTOMIZATION

- Recap Of Tanstack Table
 - Planning On Reusable Table Component
 - Introduction To Reusable Data Table
 - Creating Data Table Columns With Actions
 - Completing Reusable Data Table Component
 - Enhancing Doctor Table Columns
 - Creating Date Cell And User Data Info Cell Component For Table
 - Creating Status Badge Cell Component For Table
 - Adding Specialty Column In Doctor Columns
-

MODULE 44 / DOCTORS TABLE FEATURES & FUNCTIONALITIES

- Planning How Query Filtering Will Work From Frontend
 - Receiving Query Params From Frontend URL And Sending To API
 - Setting Up Sorting Feature In Data Table
-

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- Adding Sorting Handler Function In Doctors Table
 - Implementing Pagination Feature In Doctors Table
 - Implementing Search Feature In Doctors Table
 - Implementing Filter Feature In Doctors Table
 - Implementing Create Doctor Feature In Doctor Table
 - Implementing Edit And Delete Doctor Feature In Doctors Table
 - Implementing View Doctor Feature And Organizing Codes In Doctors Table With Reusable Hooks
-

MODULE 45 / DOCTOR SCHEDULING & APPOINTMENT FEATURES

- Creating Schedules Management Features For Admin Dashboard
- Creating Doctor Schedules Management Features For Doctors Dashboard
- Implementing Consultation Page For Public View Of Doctors Part-1
- Implementing Consultation Page For Public View Of Doctors Part-2
- Implementing Book Appointment And Payment Feature Part-1
- Implementing Book Appointment And Payment Feature Part-2

07

BE A CLOUD
CONTAINER
CRAFTSMAN

MODULE 46 / INTRODUCTION TO DOCKER

- Get Started With Docker
 - What is Docker & Why?
 - Virtual Machines vs. Docker Containers
 - Where We Use Docker
 - How Docker Works in Different OS
 - Docker Installation and Setup (Windows)
 - Docker Installation and Setup (Mac)
 - Docker Installation and Setup (Linux)
 - Configuring VS Code & Creating the First Container
-

MODULE 47 / DOCKER IMAGE AND CONTAINER

- What is Images & Containers
 - Understanding Image Types and Using Pre-built Images
 - Writing Our First Dockerfile
 - Building an Image & Running Container Based On Our Image
 - Deep Dive Into Docker Images
 - Managing Images and Practice Mostly Used Commands
 - Managing Containers and Practice Mostly Used Commands
-

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- Docker Container Restart Policies
 - Pushing Docker Image to DockerHub
 - Pulling Docker Image from DockerHub
-

MODULE 48 / DOCKER VOLUMES AND DATA MANAGEMENT

- Why Data Management Matters in Docker
 - Understanding Data Categories
 - The Writable Container Layer Explained
 - Understanding Docker Volume
 - Managing Volumes with the Docker CLI
 - Hands-on Docker Bind Mounts
 - Solving a Classic Problem Combining Volumes and Bind Mounts
 - Environment Variables and .env Files
 - The .dockerignore File
 - Volumes vs Bind Mounts Decision Guide
-

MODULE 49 / DOCKER NETWORKING AND CROSS-CONTAINER COMMUNICATION

- Container Communication Types
 - Container to WWW
 - Container to Host
 - Container to Container
 - App Communication Analysis
 - Container to Web Setup
 - Container to Host Setup
 - Container to Container Solution
 - Docker Networks
 - Network Management & IP Resolution
-

MODULE 50 / BUILDING MULTI-CONTAINER APPLICATIONS

- Steps to Build Multi-Container App
 - App Analysis for Multi-Container
 - Dockerizing MongoDB
 - Dockerizing Node.js App
 - Dockerizing Next.js App
 - Running Next.js with Bind Mount
 - Persisting MongoDB Data
 - Securing MongoDB Container
-

MODULE 51 / DOCKER COMPOSE AND MAKEFILE AUTOMATION

- Introduction to Docker Compose
 - Creating First Compose File
-

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- MongoDB Service in Compose
 - Node-Express Service in Compose
 - Next.js Service in Compose
 - Utility Containers Introduction
 - Node-Express Setup with Utility Container
 - Makefile Automation
-

MODULE 52 / REVERSE PROXY AND LOAD BALANCING WITH NGINX

- Introduction to Reverse Proxy
- How Nginx Works
- Setting Up Nginx with Docker
- Reverse Proxy Configuration with Nginx
- Routing Requests to Multiple Services
- Serving Static Files with Nginx
- Load Balancing Concepts
- Load Balancing with Nginx
- Nginx Upstream Configuration
- Health Checks and Failover Basics

08

AI ESSENTIALS FOR FULL STACK DEVELOPERS

MODULE 53 / AI, ML & LLM BASICS

- Artificial intelligence and machine learning
 - Key ML concepts: supervised, unsupervised, reinforcement learning
 - Large language models (LLMs) overview
 - Prompt engineering basics
 - Token usage and optimization
 - AI limitations and hallucination
 - Real-world AI application examples
-

MODULE 54 / AI INTEGRATION WITH NODE.JS

- Connecting AI APIs
 - Managing prompt templates
 - Structured JSON responses
 - Streaming responses
 - Error handling and retries
 - Scalable AI service architecture
 - Logging and monitoring AI calls
-

MODULE 55 / LANGCHAIN FOR LLM ORCHESTRATION

- Chains and prompt templates
- Output parsers for structured responses
- Memory and conversation state management
- LangChain agents and tools
- Multi-step workflow orchestration
- Debugging chains and agent behavior

MODULE 56 / AI CHAT ASSISTANT

- Conversation history
- Streaming responses in Next.js
- System prompts and behavior control
- Role-based assistant personas
- Multi-step workflow orchestration
- Debugging chains and agent behavior

MODULE 57 / AI MEDICAL REPORT SUMMARIZATION

- Text cleaning and preprocessing
- Document chunking strategies
- Structured summary generation
- Multi-page document handling
- Extracting key medical insights
- Formatting summaries for clarity

MODULE 58 / EMBEDDINGS & SEMANTIC SEARCH

- Vector embeddings overview
- pgvector setup in PostgreSQL
- Semantic search queries
- Indexing for performance
- Combining keyword and semantic search
- Optimizing embeddings for scale

MODULE 59 / RAG SYSTEM IMPLEMENTATION

- Retrieval-Augmented Generation (RAG) basics
- Building the retrieval pipeline
- Context injection into prompts
- Reducing hallucination with source grounding
- Multi-step RAG workflows
- Evaluating retrieval relevance

MODULE 60 / LLM FINETUNING & CUSTOMIZATION

- Finetuning vs RAG approaches
-

-
- Dataset preparation in JSON format
 - Running finetune jobs via OpenAI API
 - LoRA and QLoRA overview
 - Evaluating finetuned models
 - Practical customization use cases
-

MODULE 61 / AI AUTOMATION WITH N8N

- Self-hosting n8n with Docker
- Building low-code AI workflows
- Connecting Slack, email, and CRM
- Agentic multi-step automation
- Error handling and conditional flows
- Scheduling and monitoring workflows



REWARD COURSES (BONUS)

MODULE R1 / SOFTWARE ENGINEERING ESSENTIALS

- Foundations of Software Engineering
 - Software Design & Architecture
 - Testing & Deployment Strategies
 - Quality, Maintenance and Ethics
-

MODULE R2 / BE A TESTING TITAN

- Getting Started With Unit And Integration Tests
 - Understanding Advance Mocking Technique
 - Debugging & Rest API Testing With Vitest & Supertest
-

MODULE R3 / BE AN AWS ADVENTURER

- Getting Started With Cloud Computing
 - Comprehensive S3 And IAM Guide
 - Mastering Linux Essentials: A Beginner's Guide
 - Mastering Linux Essentials: Permissions, Processes, And Utilities
 - EC2 Essentials: Build, Deploy, And Scale With AWS
 - Server Deployment With NGINX And EC2
 - Automating CI/CD With GitHub Actions
 - Hosting An Application With GitHub Actions (CI/CD)
-